# Quadratic Functions

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Solving multi-step equations

For the model $f(x) = 5x + 9$ in solving multi-step equations, evaluate $f(5)$.

Final Answer: $34$. Substitute $x = 5$: $f(5) = 5(5) + 9 = 34$.

**2** [Foundational] • Target: Solving multi-step equations

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

Final Answer: $x = 4$ and $x = 7$. Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

**3** [Foundational] • Target: Literal equations and formulas

For the model $f(x) = 6x + 3$ in literal equations and formulas, evaluate $f(2)$.

Final Answer: $15$. Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

**4** [Foundational] • Target: Literal equations and formulas

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

Final Answer: $x = 1$ and $x = 5$. Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

**5** [Foundational] • Target: Literal equations and formulas

A data set recorded for literal equations and formulas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: $21.5$. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

**6** [Foundational] • Target: Literal equations and formulas

For the model $f(x) = 6x + 3$ in literal equations and formulas, evaluate $f(3)$.

Final Answer: $21$. Substitute $x = 3$: $f(3) = 6(3) + 3 = 21$.

**7** [Foundational] • Target: Literal equations and formulas

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

Final Answer: $x = 2$ and $x = 6$. Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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**8** [Intermediate] • Target: Literal equations and formulas

For the model $f(x) = 6x + 3$ in literal equations and formulas, evaluate $f(4)$.

Final Answer: $27$. Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

**9** [Intermediate] • Target: Literal equations and formulas

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

Final Answer: $x = 3$ and $x = 7$. Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

**10** [Intermediate] • Target: Literal equations and formulas

For the model $f(x) = 6x + 3$ in literal equations and formulas, evaluate $f(5)$.

Final Answer: $33$. Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

**11** [Intermediate] • Target: Literal equations and formulas

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

**12** [Intermediate] • Target: Practical problems with equations

For the model $f(x) = 2x + 4$ in practical problems with equations, evaluate $f(2)$.

Final Answer: $8$. Substitute $x = 2$: $f(2) = 2(2) + 4 = 8$.

**13** [Intermediate] • Target: Practical problems with equations

A data set recorded for practical problems with equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: $22.5$. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by $4$ gives $22.5$.

**14** [Intermediate] • Target: Practical problems with equations

For the model $f(x) = 2x + 4$ in practical problems with equations, evaluate $f(3)$.

Final Answer: $10$. Substitute $x = 3$: $f(3) = 2(3) + 4 = 10$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Practical problems with equations • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 2x + 4$ in practical problems with equations, evaluate $f(4)$.

Final Answer: $12$. Substitute $x = 4$: $f(4) = 2(4) + 4 = 12$.

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**16** [Challenging] • Target: Practical problems with equations • Cross-Lesson: 8.01 Factoring Polynomials

For the model $f(x) = 2x + 4$ in practical problems with equations, evaluate $f(5)$.

Final Answer: $14$. Substitute $x = 5$: $f(5) = 2(5) + 4 = 14$.

**17** [Challenging] • Target: Review: Solving one-step inequalities • Cross-Lesson: 9.01 Radicals & Exponents

For the model $f(x) = 3x + 5$ in review: solving one-step inequalities, evaluate $f(2)$.

Final Answer: $11$. Substitute $x = 2$: $f(2) = 3(2) + 5 = 11$.

**18** [Challenging] • Target: Review: Solving one-step inequalities • Cross-Lesson: 1.01 Linear Equations & Lines

A data set recorded for review: solving one-step inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: $23.5$. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by $4$ gives $23.5$.

**19** [Challenging] • Target: Review: Solving one-step inequalities • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 3x + 5$ in review: solving one-step inequalities, evaluate $f(3)$.

Final Answer: $14$. Substitute $x = 3$: $f(3) = 3(3) + 5 = 14$.

**20** [Challenging] • Target: Review: Solving one-step inequalities • Cross-Lesson: 11.01 Statistics & Data Analysis

For the model $f(x) = 3x + 5$ in review: solving one-step inequalities, evaluate $f(4)$.

Final Answer: $17$. Substitute $x = 4$: $f(4) = 3(4) + 5 = 17$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Review: Solving one- and two- step equations

For the model $f(x) = 3x + 7$ in review: solving one- and two- step equations, evaluate $f(2)$.

**a.** $13$ ✓

**b.** $16$

**c.** $6$

**d.** $17$

Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

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**22** [Foundational] • Target: Review: Solving one- and two- step equations

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

**a.** $x = 2$ and $x = 8$

**b.** $x = 2$ **and** $x = 7$ ✓

**c.** $x = -2$ and $x = -7$

**d.** $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

**23** [Foundational] • Target: Review: Solving one- and two- step equations

For the model $f(x) = 3x + 7$ in review: solving one- and two- step equations, evaluate $f(5)$.

**a.** 22 ✓

**b.** 25

**c.** 15

**d.** 26

Substitute $x = 5$: $f(5) = 3(5) + 7 = 22$.

**24** [Foundational] • Target: Properties of equality

For the model $f(x) = 4x + 8$ in properties of equality, evaluate $f(2)$.

**a.** 16 ✓

**b.** 20

**c.** 8

**d.** 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

**25** [Foundational] • Target: Properties of equality

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ **and** $x = 4$ ✓

**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

**26** [Foundational] • Target: Properties of equality

For the model $f(x) = 4x + 8$ in properties of equality, evaluate $f(5)$.

**a.** 28 ✓

**b.** 32

**c.** 20

**d.** 32

Substitute $x = 5$: $f(5) = 4(5) + 8 = 28$.

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**27** [Foundational] • Target: Solving multi-step equations

For the model $f(x) = 5x + 9$ in solving multi-step equations, evaluate $f(2)$.

**a.** 19 ✓

**b.** 24

**c.** 10

**d.** 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

**28** [Foundational] • Target: Solving multi-step equations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ **and** $x = 5$ ✓

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Review: Solving one- and two- step equations

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

**a.** $x = 1$ and $x = 7$

**b.** $x = 1$ **and** $x = 6$ ✓

**c.** $x = -1$ and $x = -6$

**d.** $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

**30** [Intermediate] • Target: Review: Solving one- and two- step equations

A data set recorded for review: solving one- and two- step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** 16.5

**b.** 17.5

**c.** 18.5 ✓

**d.** 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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**31** [Intermediate] • Target: Review: Solving one- and two- step equations

For the model $f(x) = 3x + 7$ in review: solving one- and two- step equations, evaluate $f(3)$.

**a.** 16 ✓

**b.** 19

**c.** 9

**d.** 20

Substitute $x = 3$: $f(3) = 3(3) + 7 = 16$.

**32** [Intermediate] • Target: Review: Solving one- and two- step equations

For the model $f(x) = 3x + 7$ in review: solving one- and two- step equations, evaluate $f(4)$.

**a.** 19 ✓

**b.** 22

**c.** 12

**d.** 23

Substitute $x = 4$: $f(4) = 3(4) + 7 = 19$.

**33** [Intermediate] • Target: Review: Solving one- and two- step equations

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

**a.** $x = 3$ and $x = 9$

**b.** $x = 3$ **and** $x = 8$ ✓

**c.** $x = -3$ and $x = -8$

**d.** $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

**34** [Intermediate] • Target: Review: Solving one- and two- step equations

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

**a.** $x = 4$ and $x = 10$

**b.** $x = 4$ **and** $x = 9$ ✓

**c.** $x = -4$ and $x = -9$

**d.** $x = 13$ and $x = 36$

Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

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**35** [Intermediate] • Target: Properties of equality

A data set recorded for properties of equality consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5 ✓

**d.** 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by $4$ gives $19.5$.

**36** [Intermediate] • Target: Properties of equality

For the model $f(x) = 4x + 8$ in properties of equality, evaluate $f(3)$.

**a.** 20 ✓

**b.** 24

**c.** 12

**d.** 24

Substitute $x = 3$: $f(3) = 4(3) + 8 = 20$.

**37** [Intermediate] • Target: Properties of equality

For the model $f(x) = 4x + 8$ in properties of equality, evaluate $f(4)$.

**a.** 24 ✓

**b.** 28

**c.** 16

**d.** 28

Substitute $x = 4$: $f(4) = 4(4) + 8 = 24$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Properties of equality • Cross-Lesson: 1.01 Linear Equations & Lines

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

**a.** $x = 3$ and $x = 6$

**b.** $x = 3$ **and** $x = 5$ ✓

**c.** $x = -3$ and $x = -5$

**d.** $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

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**39** [Challenging] • Target: Properties of equality • Cross-Lesson: 10.01 Rational Expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

**a.** $x = 4$ and $x = 7$

**b.** $x = 4$ **and** $x = 6$ ✓

**c.** $x = -4$ and $x = -6$

**d.** $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

**40** [Challenging] • Target: Solving multi-step equations • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ **and** $x = 4$ ✓

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

**41** [Challenging] • Target: Solving multi-step equations • Cross-Lesson: 3.01 Exponents & Polynomials

For the model $f(x) = 5x + 9$ in solving multi-step equations, evaluate $f(3)$.

**a.** 24 ✓

**b.** 29

**c.** 15

**d.** 28

Substitute $x = 3$: $f(3) = 5(3) + 9 = 24$.

**42** [Challenging] • Target: Solving multi-step equations • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 5x + 9$ in solving multi-step equations, evaluate $f(4)$.

**a.** 29 ✓

**b.** 34

**c.** 20

**d.** 33

Substitute $x = 4$: $f(4) = 5(4) + 9 = 29$.

**43** [Challenging] • Target: Solving multi-step equations • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ **and** $x = 6$ ✓

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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**44** [Challenging] • Target: Literal equations and formulas • Cross-Lesson: 6.01 Systems & Inequalities

A data set recorded for literal equations and formulas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

**a.** 19.5

**b.** 20.5

**c.** 21.5 ✓

**d.** 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by $4$ gives $21.5$.

**45** [Challenging] • Target: Practical problems with equations • Cross-Lesson: 7.01 Sequences & Series

A data set recorded for practical problems with equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

**a.** 20.5

**b.** 21.5

**c.** 22.5 ✓

**d.** 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by $4$ gives $22.5$.

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